

## ▽ ATTENTION SEEKER

The male's throat pouch expands into a red balloon as he sits on a treetop nest, displaying to female frigatebirds flying overhead.



inflated pouch

# Great frigatebird

*Fregata minor*



long tail used as rudder when flying

The great frigatebird looks almost prehistoric, with its long beak, forked tail, and huge, pointed, angular wings set in an inverted "W." It has the largest wing area to body mass, or the lowest wing-loading, of any bird. This, together with the large tail spread wide or closed to a single point, makes the frigatebird extremely stable as it soars effortlessly for hours, as well as supremely agile while dashing to grab flying fish or chase other birds. It also steals food from seabirds, especially boobies, harassing them until they regurgitate. The frigatebird flies over the sea, but avoids landing on water because it lacks fully waterproof plumage. It nests in trees on remote tropical islands in the Pacific, Indian, and South Atlantic oceans. Pairs take turns to incubate one egg for a period of 55 days.

- ↔ 34–41in (85–105 cm)
- ⚖ 2–3 lb (1–1.5 kg)
- ⊗ Common
- 🐟 Fish, squid, seabird chicks
- 🏠

Tropical Pacific, South Atlantic, and Indian Oceans

Frigatebirds **rarely land during the day**, except when breeding

# Galapagos penguin

*Spheniscus mendiculus*

The only penguin that lives north of the Equator, the Galapagos penguin breeds mostly on the Fernandina and Isabela islands of the Galapagos archipelago. Swimming in the Cromwell Current by day, they exploit small schooling fish such as mullets and sardines—the cold ocean current provides a higher nutrient content than warm, tropical waters. The penguins visit land at night, when it is both cooler and safer for flightless birds. Like Antarctic penguins, they can flap their tiny wings to lose excess body heat.

## Rare bird

The Galapagos penguin is one of the world's rarest penguins, with fewer than 1,000 breeding pairs. Their breeding season is regulated by the availability of food. Climatic events such as El Niño cycles warm the waters around the islands, reducing fish numbers as they depart for cooler waters. The resulting food shortage makes the penguins skip an entire breeding season.

Galapagos penguins are also threatened by pollution, intensive fishing, and predators such as cats and dogs that human settlers have brought to the islands.

- ↔ 21in (53cm)
- ⚖ 4–5lb (1.8–2.3kg)
- ⊗ Endangered
- 🐟 Mullet, sardines
- 🏠



Galapagos Islands

## ▷ PARTNERS FOR LIFE

Galapagos penguins mate for life. The female lays one or two eggs in deep rock crevices to keep them cool. Both parents take turns incubating them for a period of 38–40 days.

